BM1500 Week 7 HW1 Hypothesis

For this assignment I chose HW1 which asks for ~~a~~ a Susceptible-Infectious-Recovered (SIR) compartmental model to be implemented and extended. The change in population within each compartment is defined by the given ODEs and shaped by the transmission Rate (β) and Recovery Rate (γ), thus ~~we~~ with initial conditions ~~we~~ of the Susceptible (S), Ifected (I), and Recovered (R) populations we can extrapoliate overtime.

Part B: Initially I will grow as $\frac{\beta N}{\gamma} > 0$, and using the ODE $\frac{dI}{dt} = (\beta S - \gamma)I$ the initial infections per day is ~~at~~ 0.149. At the peak $\frac{dS}{dt} = \frac{dR}{dt}$ which gives $S = \gamma/\beta = 800$. Thus to estimate the day and population of I using ~~initial~~ 0.149 Rate for I would would double ~4.5 days, thus to go from 5 to 640 for I would be 32 days or 7 doublings. Around 600-700 people for I at the peak since 1,195 possible between I & R with slightly more being in I

Part D: Adding the exposed compartment gives a delay of $\frac{1}{\sigma}$ before changing to I thus the first peak will be later more days and flatter than in Part B. There will be multiple waves ~~in~~ now that become smaller, this is caused by the Birth Rate ~~[illegible]~~ adding to the S population leading to increase in I.

Part E: I hypothesize that β will have a greater on the peak Infected population. Simply Recovery Rate is constant, while transmission Rate is exponential as one person can infect many others. Higher β will give an earlier and larger peak and vice versa.